MicroStitch SDK to MFC Integration

These are the complete steps to perform **inside the MeasurePro MFC/Visual Studio project**. No changes to the SDK are needed.

1. Extract the SDK

Extract:

```
MicroStitchSDK_Handoff.zip
```

For example:

```
D:\MicroStitchSDK_Handoff\
```

It should contain:

```
MicroStitchSDK_Handoff/
|-- README.md
|-- CMakeLists.txt
|-- include/
|   `-- microstitch/
|       `-- stitcher.h
|-- src/
|   |-- registration.h
|   |-- registration.cpp
|   |-- mosaic.h
|   |-- mosaic.cpp
|   |-- mosaic_renderer.h
|   |-- mosaic_renderer.cpp
|   |-- stitcher.h
|   `-- stitcher.cpp
|-- lib/
|   `-- MicroStitch.lib
`-- bin/
    `-- MicroStitch.dll
```

2. Set MFC project to x64 Release

In Visual Studio: **Build > Configuration Manager**. Set:

```
Configuration: Release
Platform: x64
```

The MicroStitch SDK is an **x64 Release** build.

3. Add the MicroStitch header path

Right-click **MeasurePro > Properties**. Select Configuration: Release, Platform: x64. Go to **C/C++ > General > Additional Include Directories** and add:

```
D:\MicroStitchSDK_Handoff\include
```

Use whatever actual location the SDK was extracted to.

4. Add OpenCV 3.4.11 include path

In the same place, add:

```
C:\tools\opencv3411\include
```

So the include directories contain both:

```
D:\MicroStitchSDK_Handoff\include
C:\tools\opencv3411\include
```

5. Add MicroStitch library path

Go to **Linker > General > Additional Library Directories** and add:

```
D:\MicroStitchSDK_Handoff\lib
```

6. Add OpenCV library path

Also add:

```
C:\tools\opencv3411\lib
```

So:

```
D:\MicroStitchSDK_Handoff\lib
C:\tools\opencv3411\lib
```

7. Add MicroStitch.lib

Go to **Linker > Input > Additional Dependencies** and add:

```
MicroStitch.lib
```

This step is particularly important because the current `LNK2001` error means the linker is currently not resolving the MicroStitch implementation.

8. Add the required OpenCV libraries

Because the MicroStitch DLL was built against OpenCV 3.4.11, the MFC project needs to link against the compatible OpenCV libraries as appropriate for the existing MeasurePro OpenCV setup. The MicroStitch build used:

```
opencv_core3411.lib
opencv_imgproc3411.lib
opencv_features2d3411.lib
opencv_calib3d3411.lib
opencv_flann3411.lib
opencv_imgcodecs3411.lib
```

If MeasurePro already has these OpenCV libraries configured, **do not add duplicate entries**.

9. Include the MicroStitch header

In the MFC .cpp file where stitching is used:

```
#include <opencv2/opencv.hpp>
#include <microstitch/stitcher.h>
```

10. Create one persistent Stitcher object

Do **not** create a new Stitcher for every frame. For example, in the MFC dialog/class header:

```
microstitch::Stitcher m_stitcher;
```

Or:

```
microstitch::Stitcher m_stitcher(0.05, 0.35, 150.0);
```

The object should live for the duration of the stitching session/application.

11. Process the microscope frames

Where MeasurePro already receives the microscope `cv::Mat`:

```
bool accepted = m_stitcher.processFrame(frame);
```

Then:

```
if (m_stitcher.hasMosaic())
{
    cv::Mat mosaic = m_stitcher.getMosaic();

    // Display mosaic in the MFC mosaic window
}
```

The existing camera frame can continue going to the existing camera display.

12. Do NOT use the old API

Do **not** use:

```
Stitcher_Create()
Stitcher_ProcessFrame()
Stitcher_GetMosaic()
Stitcher_Reset()
Stitcher_Destroy()
```

Those were from the previous SDK. The final SDK uses:

```
m_stitcher.processFrame(frame);
m_stitcher.hasMosaic();
m_stitcher.getMosaic();
m_stitcher.reset();
```

13. Put runtime DLLs beside the MeasurePro EXE

At runtime, Windows needs to find `MicroStitch.dll` and the OpenCV 3.4.11 runtime DLLs required by MicroStitch:

```
MicroStitch.dll
opencv_core3411.dll
opencv_imgproc3411.dll
opencv_features2d3411.dll
opencv_calib3d3411.dll
```

The simplest arrangement is to put them in the same directory as:

```
MeasurePro.exe
```

14. Build the project

Do **Build > Clean Solution**, then **Build > Rebuild Solution**. Make sure the build configuration is:

```
Release | x64
```

15. If you still get LNK2001

Errors such as the following indicate that the header was found but the implementation wasn't linked:

```
unresolved external symbol
__declspec(dllimport) microstitch::Stitcher::Stitcher()
```

Check these three things first:

A. Linker > Input

Must contain:

```
MicroStitch.lib
```

B. Linker > General

Must contain the folder containing MicroStitch.lib, for example:

```
D:\MicroStitchSDK_Handoff\lib
```

C. Configuration

Both settings must be under **Release** and **x64**, not Debug/Win32.

As a quick diagnostic, you can also temporarily put this after the `#include <microstitch/stitcher.h>` line:

```
#pragma comment(lib, "MicroStitch.lib")
```

If that fixes the linker error, the problem was definitely the Visual Studio library configuration.

16. Save functionality

The SDK doesn't save files itself. The MFC application handles it. For example:

```
cv::Mat mosaic = m_stitcher.getMosaic();

if (!mosaic.empty())
{
    cv::imwrite("mosaic.png", mosaic);
}
```

So the existing **Save** button in MeasurePro can call this.

17. Reset functionality

For a new slide/session:

```
m_stitcher.reset();
```

The next frame becomes the starting point of the new mosaic.

18. Final architecture

The finished MeasurePro application should effectively work like:

```
        MeasurePro MFC
              |
      Microscope Camera
              |
          cv::Mat
              |
              v
   m_stitcher.processFrame()
              |
              v
       MicroStitch.dll
              |
        OpenCV 3.4.11
              |
              v
    m_stitcher.getMosaic()
              |
              v
          cv::Mat
              |
              v
      MFC Mosaic Window
```

In short: four major things

1. **Add include + OpenCV include paths**
2. **Add lib + OpenCV library paths and MicroStitch.lib**
3. **Put the required DLLs beside MeasurePro.exe**
4. **Use the C++ microstitch::Stitcher API instead of the old C API**

The SDK itself does **not** need to be rebuilt or modified to fix the `LNK2001` error.